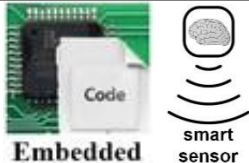
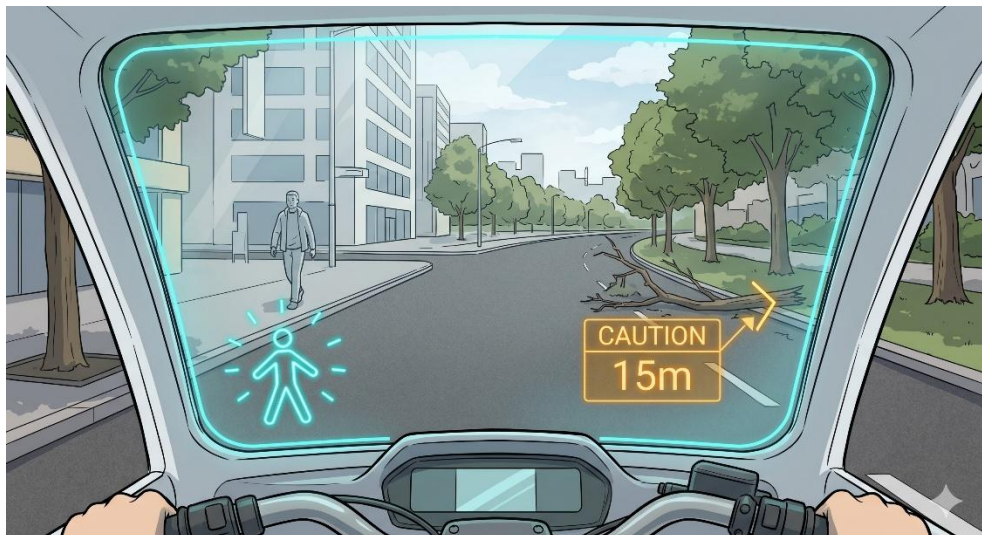


ID:	SW-7	
SUPERVISOR:	Simon Winberg Co-supervisor: Sampath Jayalath	
TITLE:	Development of an Embedded HPEC Telemetry Unit and Windshield HUD for the TUKZIE Rev 0 Platform	
DESCRIPTION:	The TUKZIE Flagship Programme aims to engineer a secure, energy-aware, autonomous-ready urban mobility vehicle. Central to Tukzie Rev 0 is vehicle characterization and creation of a baseline telemetry architecture that feeds directly into the system's foundational mechanical and electrical digital twins. This project involves developing a High-Performance Embedded Computing (HPEC) telemetry unit and driver assistance interface for the TUKZIE platform, illustrated in Fig. 1 below.  Fig. 1: Trike HUD showing 'augmented reality' projections on the windshield. This system carries a dual mandate. First, it will perform real-time edge processing on an array of integrated vehicle and environmental sensors, for monitoring powertrain performance, ride smoothness (via IMU data), microclimate profiles, and localized air quality metrics to detect environmental anomalies (e.g., plastic burning). Second, the unit will drive a dynamic graphical user interface projected as a Heads-Up Display (HUD) onto a custom handlebar-mounted front windshield visor. This HUD will map critical telemetry (speed, battery State-of-Charge, fleet notifications) and render active driver alerts based on localized sensor feeds. Telemetry will be logged locally and transmitted via a cellular module (4G/LTE) to a central database for route optimization.	
DELIVERABLES:	• Deliverable 1: An integrated HPEC telemetry hardware module housing the environmental, air quality, and IMU sensor payload, alongside an optical HUD projection subsystem. • Deliverable 2: A deterministic multi-threaded software architecture executing real-time sensor fusion, signal processing for ride-smoothness metrics, and low-latency GUI rendering for the HUD. • Deliverable 3: A structured technical report validating data ingestion	

	accuracy, edge processing latency, and successful remote data synchronization with the central database.
SKILLS/ REQUIREMENTS:	• Proficiency in embedded C/C++ or Python within real-time operating systems (e.g., FreeRTOS) or Embedded Linux environments. • Hands-on experience with sensor protocols (I2C, SPI, UART) and graphical interface development tools. • Familiarity with cellular IoT modules (MQTT/HTTP over 4G/LTE-M).
GA 1: Problem solving: Identify, formulate, analyse and solve complex* engineering problems creatively and innovatively	The student must architect an embedded pipeline that cleanly executes parallel execution paths: high-frequency sensor polling, compute-heavy signal processing algorithms for ride characterization, graphical HUD rendering, and networking stacks, ensuring no process blocking occurs.
GA 4**: Investigations, experiments and analysis: Demonstrate competence to design and conduct investigations and experiments.	The student will construct field testing procedures to evaluate the accuracy and reliability of the edge-sensor array under real driving conditions. They will compile and systematically analyze logged spatial and electrical datasets to successfully isolate true road roughness from nominal engine vibrations.
GA 5: Use of engineering tools: <i>Demonstrate competence to create, select and apply and recognise limitations of appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems</i>	The student will deploy advanced embedded profiling utilities, hardware debuggers, and cloud telemetry instrumentation tools. They will be required to analyze and operate within the explicit bounds of the processing hardware, the available electrical draw from the main 72V battery, and wireless bandwidth limits.
EXTRA INFORMATION:	[1] What the cargo trike looks like https://www.samotorcycles.co.za/monster-cargo-e-trike-2200w-lithium [2] Recommended watch and potential system to base design on: https://www.youtube.com/watch?v=aI99zUIezX0 Student is required to prepare a thorough ethics proposal emphasizing the safety of testing, such as in unused parking area or permission to use a field, where the system can be tested on an actual trike vehicle. Furthermore, the projector device (or custom windshield) needs to be easily removable so as to ensure the vehicle can quickly be adjusted to a standard setup without any obstructions or interference from devices used in this project.
BROAD Research Area:	Embedded Systems / Edge Computing
Project suitable for ME/ ECE/EE/ALL?	ALL

Ethics clearance questionnaire

		Yes	No
Q1	Does this project involve data collection	X	
Q2	Does this project involve utilizing a third-party data set		X
Q3	Does this project utilize machine learning (ML) or artificial intelligence (AI)?		X
Q4	Does it exceed the minimum risk defined here: Link [Answer is No here if your project does not utilize ML and AI]		X
Q5	Does this project involve external parties, funders, etc	X	

Answer the following questions if you answer "Yes" to any of the above questions.

If the answer is "Yes" to **Q1**, please answer the following questions:

		Yes	No
Q6	Are there humans or animals directly involved in the data collection process or contains any identification information		X

Notes on Q3: At present, machine learning (ML) is not planned for use in this version (as in first version) of this proposed system. However, should the student doing project be making good headway and consider the baseline system is sufficiently progress to merit experimenting with ML, then a revised ethics application will be prepared a submitted.

Note on Q5: The extent to which external parties may be involved, the availability of permission of using a trike will be discussed and planned as part of the project. While much of the designs and development methods are planned to be potentially shared or available in open-access repositories, these aspects are not yet finalized or decided. If there is need for MOUs, this will be clarified; the use of the trike and sharing detailed information about it (e.g. performance information) that may or may not be beneficial to the manufacture needs to be handled appropriately and in agreement with the relevant stakeholders concerned.